ARYA V P

LinkedIn | GitHub
DOB – 05/10/2005
aryavp.mec@gmail.com
+91 9778589447

SKILLS & INTERESTS

- **Technical Skills:** C, Python, 8051, Arduino, ESP32, Proteus, Fusion 360, Onshape, Embedded Systems, OpenCV, MediaPipe, PCB Design
- **Soft Skills:** Communication, Problem Solving, Critical Thinking, Teamwork, Leadership, Adaptability, Attention to Detail, Time Management, Organisational Skills
- **Interests:** Embedded Systems, IoT Development, Robotics, Automation Systems, Control Systems, 3D Design and CAD Modelling

EDUCATION

- **Govt. Model Engineering College** — **8.65** | 2027
  KTU, B.Tech in Electronics and Communication Engineering
- **Govt. Girls Higher Secondary School** — **99.17%** | 2023
  State, 12th
- **St. Mary's HS for Girls** — **100%** | 2021
  State, 10th

WORK EXPERIENCE

- **iHUB ROBOTICS** — *2 Weeks*
  Intern
  Technologies Used: *Arduino, ESP32, Embedded C, Sensors & Actuators*
  Completed a two-week internship gaining hands-on experience in robotics and IoT systems, working with embedded hardware, sensor integration and automation while applying concepts through practical project-based implementation.

PROJECTS

- **Gesture-Controlled Robotic Arm** — **Team Size: 4**
  Team Lead — *1 Month*
  Technologies Used: *Python, Media Pipe, Arduino, Servo Motors, Computer Vision*
  Developed a gesture-controlled robotic arm by integrating computer vision algorithms with hardware actuation, enabling real-time gesture recognition and motor control through seamless hardware-software interaction.
- **Maze Solving Robot** — **Team Size: 4**
  Team Lead — *2 Weeks*
  Technologies Used: *Arduino, IR Sensors, Embedded C, Motor Drivers*
  Designed an autonomous maze-solving robot using sensor-based decision making and the Left-Hand Rule maze solving algorithm, implementing embedded control logic for path navigation and obstacle detection.
- **ESP32-Based Home Automation System** — **Team Size: 1**
  Team Lead — *2 Weeks*
  Technologies Used: *ESP32, Embedded C, Sensors, Relays, IoT Platform*
  Designed and implemented a wireless home automation system enabling remote control of electrical appliances through ESP32, integrating sensors, relay modules and embedded firmware for reliable device automation.
- **Quantum Random Number Generator (QRNG) with AES Encryption** — **Team Size: 4**
  Team Member — *1 Month*
  Technologies Used: *Arduino/ESP32, Analog Noise Circuit, Embedded C, Python, AES Encryption*
  Designed and developed a hardware-based Quantum Random Number Generator utilizing noise from a semiconductor source to generate true random bits. Implemented real-time data acquisition using a microcontroller and integrated AES encryption for secure key generation and communication.

COURSES AND CERTIFICATIONS

- Certified in **Introduction to the World of VLSI** by **Udemy**, gaining foundational knowledge of VLSI design concepts, semiconductor devices and basic chip design principles.

ACHIEVEMENTS AND ACTIVITIES

- Achieved **1st Prize** in **MEC Labs**, conducted as part of **Excel 2024**, the annual techno-managerial fest of **Govt. Model Engineering College**.
- Achieved **1st Prize** in **Reverso**, conducted as part of **Excel 2025**, the annual techno-managerial fest of **Govt. Model Engineering College**.
- Secured **3rd Prize** in **Omega**, a Maze Solving Robot Competition organised by **IEEE MEC SB**.
- Participated in **Tharang 2025**, a technical exhibition conducted by **IHRD**.
- Volunteered and mentored students in **REMAP 2024**, a hardware hackathon conducted as part of **Excel 2024**, the annual techno-managerial fest of **Govt. Model Engineering College**.
- Worked as an **IDEALAB Trainee 2025**, gaining hands-on experience in hardware development and prototyping.
- **Hobbies:** Painting, Taekwondo, Dancing, Learning Musical Instruments.

REFERENCES

- **Prof. Dr. Jayachandran E S**, Principal, Govt. Model Engineering College, Kochi, Email ID: **principal@mec.ac.in**
- **Prof. Dr. Jobymol Jacob**, HOD, Electronics and Communication Engineering, Govt. Model Engineering College, Kochi, Email ID: **hodec@mec.ac.in**